

A Bayesian Population-Dynamics Model for Clonal T Cells across Tissue and Phenotype in Glioblastoma

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1 Setup

We observe annotated T cells from multiple patients, anatomical sites, and study timepoints. Each observed cell carries five labels: a patient, a TCR clonotype, a study timepoint, an anatomical site, and a discrete T-cell phenotype state.

The data are represented as aggregate clone-by-timepoint-by-site-by-phenotype counts, from which we construct clone-level observations across forward study steps.

1.1 Indices

We use the following indices throughout.

- Patients: $i = 1, \dots, I$, with $I = 6$.
- TCR clonotypes within patient i : $c = 1, \dots, C_i$.
- Shared study timepoints: $r \in \{1, \dots, R\}$, ordered as $\tau_1 < \tau_2 < \dots < \tau_R$. The timepoint labels are comparable across patients. Currently $R = 6$; the cohort is being extended with additional timepoints.
- Anatomical site: $s \in \mathcal{S}$, with $\mathcal{S} = \{\text{PBMC}, \text{CSF}, \text{Tumor}\}$.
- Annotated T-cell phenotype state: $k \in \{1, \dots, K\}$. In the current application $K = 11$, and all phenotypes share a single phenotype space.

1.2 Observed count tensor

For each patient i , clonotype c , timepoint r , site s , and phenotype k , let

$$Y_{icrsk} \in \mathbb{Z}_{\geq 0}$$

be the observed number of cells assigned to that combination: how many cells from patient i , clonotype c , and timepoint r were sampled from site s and assigned phenotype k . Collecting the site and phenotype axes,

$$Y_{icr} \in \mathbb{Z}_{\geq 0}^{|\mathcal{S}| \times K}$$

is the tissue-by-phenotype count tensor for clonotype c in patient i at timepoint r .

Each tissue sample also carries a sequencing depth and a missingness flag. For patient i , timepoint r , and tissue s , let

$$d_{i,r,s} \in \mathbb{Z}_{\geq 0}$$

be the per-sample sequencing depth: the number of TCR⁺ T cells in that (patient, timepoint, tissue) sample. It is the Poisson exposure used below. Let

$$m_{i,r,s} \in \{0, 1\}$$

be the missingness indicator, equal to 1 when that sample was profiled and 0 otherwise.

1.3 Joint tissue–phenotype state space

The model operates on the joint tissue–phenotype state space

$$\mathcal{Z} = \mathcal{S} \times \{1, \dots, K\}, \quad z = (s, k),$$

where s is the tissue and k is the phenotype. Since $|\mathcal{S}| = 3$ and $K = 11$, the number of joint states is

$$L = |\mathcal{Z}| = |\mathcal{S}|K = 33.$$

The model therefore tracks clone counts across a 33-state tissue–phenotype space. When describing transitions we write the source state as $z = (a, u)$ and the destination state as $z' = (b, v)$, where $a, b \in \mathcal{S}$ are source and destination tissues and $u, v \in \{1, \dots, K\}$ are source and destination phenotypes.

1.4 Valid forward steps

Forward steps are defined per patient. For patient i , let $\mathcal{R}_i \subseteq \{1, \dots, R - 1\}$ be the set of source timepoints at which patient i has a valid forward step. We index a step by its source timepoint $r \in \mathcal{R}_i$; the step runs from timepoint r to the next sampled timepoint, written $r + 1$. Thus r is not a matrix index—it simply labels a forward step within patient i by where it starts.

Remark (Skipped timepoints). Writing the destination as $r + 1$ assumes consecutive sampling. When a patient is missing a timepoint—so that, say, T_1 and T_3 are paired— $r + 1$ should be read as the *next available* timepoint for that patient. Which timepoints are paired is a data-preprocessing detail and does not change anything below.

Each $r \in \mathcal{R}_i$ is one observed forward step. Steps such as $T_1 \rightarrow T_2$, $T_2 \rightarrow T_3$, and $T_3 \rightarrow T_4$ are treated as repeated observations of the *same* forward process: the mean matrix is shared across all patients and steps, and the indices i and r merely organize observations rather than selecting a separate matrix.

1.5 Clone-level observations within a step

Fix a patient i , a valid step $r \in \mathcal{R}_i$, and a clonotype c observed for that patient. The source and destination count vectors over the joint state space are read directly off the count tensor at timepoints r and $r + 1$,

$$x_{irc}(s, k) = Y_{i c r s k}, \quad y_{irc}(s, k) = Y_{i c (r+1) s k}, \quad (s, k) \in \mathcal{Z},$$

so $x_{irc} \in \mathbb{Z}_{\geq 0}^L$ and $y_{irc} \in \mathbb{Z}_{\geq 0}^L$.

The source enters the model as its depth-normalized composition. The *depth-rescaled source*

$$\tilde{x}_{irc}(z) = \frac{x_{irc}(z)}{d_{ir,s}^{\text{src}}}, \quad s = \text{tissue}(z), \quad d_{ir,s}^{\text{src}} = d_{i,r,s},$$

is the fraction of the source sample's TCR⁺ T cells in bin z (tissue $s = \text{tissue}(z)$). The per-sample depth d is retained rather than normalized away, so that an observed destination zero can be read as contraction or clonal loss where depth was high versus undersampling where depth was low—the separation the depth offset provides.

We keep every clonotype with at least one cell at the source timepoint r ,

$$C_{ir} = \{c \in \{1, \dots, C_i\} : N_{irc}^{\text{src}} > 0\}, \quad N_{irc}^{\text{src}} = \sum_{z \in \mathcal{Z}} x_{irc}(z).$$

A clone that is absent at the destination timepoint is kept: its absence is informative—contraction or clonal loss—not a reason to drop it. This gives the data hierarchy

$$i \longrightarrow r \in \mathcal{R}_i \longrightarrow c \in C_{ir},$$

patients contain valid forward steps, and each forward step has a retained set of clonotypes present at the source timepoint. A *clone-level forward observation* is the triple (i, r, c) ; for compact notation we write

$$j \equiv (i, r, c), \quad x_j = x_{irc}, \quad \tilde{x}_j = \tilde{x}_{irc}, \quad y_j = y_{irc}.$$

Each destination state z' carries the destination depth and missingness flag of its tissue $s' = \text{tissue}(z')$: $d_{j,s'} = d_{i,r+1,s'}$ and $m_{j,s'} = m_{i,r+1,s'}$.

1.6 Two kinds of destination zero

Destination zeros are part of the likelihood specification, and the two kinds are treated differently.

1. **Missing sample** ($m_{j,s'} = 0$). The destination states in tissue s' were not profiled at the destination timepoint. They are excluded from the likelihood entirely—masked out, never entered as a zero count.
2. **Depth-zero** ($m_{j,s'} = 1$, clone not captured). A genuine observed zero, modeled natively by the Poisson likelihood below: a small mean $d_{j,s'} \mu_j(z')$ emits zero with positive probability while $\mu_j(z') > 0$. No zero-inflation component is added.

The depth-zero mechanism is what permits a clone to be absent at one step and present at the next.

1.7 Setup summary

The setup produces the clone-level forward observations

$$\mathcal{D} = \{(\tilde{x}_j, y_j, d_j, m_j, i_j, r_j, c_j)\}_{j=1}^J, \quad j \equiv (i_j, r_j, c_j),$$

each carrying a depth-rescaled source vector, a destination count vector, the per-tissue destination depths $d_j = (d_{j,s'})_{s' \in \mathcal{S}}$ and missingness flags $m_j = (m_{j,s'})_{s' \in \mathcal{S}}$, and a patient / step / clonotype identity. The count vectors live in the joint tissue–phenotype state space $\mathcal{Z} = \mathcal{S} \times \{1, \dots, K\}$, with $|\mathcal{Z}| = 33$.

1.8 Notation summary

Table 1 collects the core modeling symbols.

Symbol	Meaning
$i = 1, \dots, I$	patient index
$c = 1, \dots, C_i$	TCR clonotype within patient i
$r \in \{1, \dots, R\}$	shared study timepoint, $\tau_1 < \dots < \tau_R$
$s \in \mathcal{S}$	anatomical site, $\mathcal{S} = \{\text{PBMC}, \text{CSF}, \text{Tumor}\}$
$k \in \{1, \dots, K\}$	phenotype state ($K = 11$)
$z = (s, k) \in \mathcal{Z}$	joint tissue–phenotype state ($L = \mathcal{Z} = 33$)
Y_{icrsk}	observed count tensor
$d_{i,r,s}$	per-sample sequencing depth (Poisson exposure)
$m_{i,r,s} \in \{0, 1\}$	sample missingness indicator
$\mathcal{R}_i$	valid forward steps for patient i ($\subseteq \{1, \dots, R - 1\}$)
$r \in \mathcal{R}_i$	forward step, from source timepoint r to $r + 1$
C_{ir}	clonotypes present at the source timepoint of step r
$j = (i_j, r_j, c_j)$	clone-level observation, $j = 1, \dots, J$
$x_j, y_j \in \mathbb{Z}_{\geq 0}^L$	source / destination count vectors
$\tilde{x}_j$	depth-rescaled source, $\tilde{x}_j(z) = x_j(z) / d_{j,s}^{\text{src}}$
$M \in \mathbb{R}_{\geq 0}^{L \times L}$	non-negative mean matrix (free row sums)
$\mu_j = \tilde{x}_j M \in \mathbb{R}_{\geq 0}^L$	destination intensity vector
$T \in \mathbb{R}^{L \times L}$	recovered transition matrix, $M = \text{diag}(g) T$, row-stochastic
$g_z = \sum_{z'} M_{zz'}$	per-state expansion factor
a_z, b_z	Gamma prior shape / rate for row z
$w_{jzz'}$	backward attribution probability
$\rho^a(u)$	empirical source-cell phenotype distribution in tissue a (pooled)
$\theta(z)$	clone source composition, source cells as a distribution over states
$\mathcal{D}$	full dataset

Table 1: Core modeling notation.

2 Joint Tissue–Phenotype Mean Matrix

The central latent object is a single joint tissue–phenotype *mean matrix*

$$M \in \mathbb{R}_{\geq 0}^{L \times L}, \quad L = 33,$$

whose rows are source states and whose columns are destination states. For source state $z = (a, u)$ and destination state $z' = (b, v)$, the entry $M_{zz'}$ is the expected number of destination- z' cells produced per source- z cell over one forward step:

$$M_{zz'} = M_{(a,u),(b,v)} = \mathbb{E}[\text{cells in state } (b, v) \text{ at } r + 1 \text{ per cell in state } (a, u) \text{ at } r].$$

M differs from a transition matrix only in that its rows are not constrained to sum to one (Section 2.1); the row sum is freed to carry expansion.

2.1 Non-negative rows and expansion factors

M has non-negative entries, $M_{zz'} \geq 0$, and *free* row sums—rows are not constrained to sum to one. The row sum

$$g_z = \sum_{z' \in \mathcal{Z}} M_{zz'}$$

is the net expansion factor (per step) of source state z : $g_z > 1$ is net expansion of that state, $g_z < 1$ net contraction or clonal loss.

Writing $g_z = \sum_{z' \in \mathcal{Z}} M_{zz'}$ and, wherever $g_z > 0$, $T_{zz'} = M_{zz'} / g_z$, the matrix factors uniquely as $M = \text{diag}(g) T$ with T row-stochastic. The two factors separate the processes: T is the transition matrix—the destination distribution conditional on a source cell having a descendant—and g is the per-state expansion or contraction a transition matrix alone discards. A conservative (row-stochastic) model is the special case $g_z \equiv 1$.

Two caveats: $T_{z\cdot}$ is defined only where $g_z > 0$; and the recovered T is the destination conditional on a descendant, coinciding with a conservative transition matrix only when expansion is uniform across states.

2.2 Destination intensity vector

For observation j with depth-rescaled source $\tilde{x}_j$, the destination *intensity vector* is the push-forward of $\tilde{x}_j$ through M :

Destination intensity

$$\mu_j = \tilde{x}_j M \in \mathbb{R}_{\geq 0}^L, \quad \text{equivalently} \quad \mu_j(z') = \sum_{z \in \mathcal{Z}} \tilde{x}_j(z) M_{zz'}.$$

The expected destination count in state z' scales the intensity by the destination depth of its tissue,

$$\mathbb{E}[y_j(z') \mid M] = d_{j,s'} \mu_j(z'), \quad s' = \text{tissue}(z').$$

There is no simplex constraint on μ_j and no conditioning on a destination total.

2.3 Role of forward steps

The same matrix M is used for every patient i and step $r \in \mathcal{R}_i$. The pair (i, r) selects which source and destination vectors to build, not a different matrix; all steps are repeated observations of a common one-step forward process.

3 Bayesian Treatment of the Mean Matrix

The matrix M is a latent non-negative kernel of per-cell destination intensities over the joint tissue–phenotype state space.

3.1 Forward generative reading

Read forward, the model is a Poisson superposition (independent Poissons sum to a Poisson with summed rates). Treating the depth-rescaled source as a population with $\tilde{x}_j(z)$ cells in each state z , each source cell in state z independently emits a Poisson-distributed number of cells into each destination state z' with mean $M_{zz'}$. Summing these emissions over all source states and scaling by the destination depth, the destination- z' count is, by Poisson superposition,

$$y_j(z') \mid M \sim \text{Poisson}\left(d_{j,s'} \sum_{z \in \mathcal{Z}} \tilde{x}_j(z) M_{zz'}\right) = \text{Poisson}(d_{j,s'} \mu_j(z')).$$

3.2 Likelihood for destination counts

Observation likelihood

$$y_j(z') \sim \text{Poisson}(d_{j,s'} (\tilde{x}_j M)(z')), \quad s' = \text{tissue}(z'),$$

evaluated only at destination states z' whose tissue is non-missing ($m_{j,s'} = 1$). The latent matrix enters the likelihood only through the intensity $\mu_j = \tilde{x}_j M$.

3.3 Prior over mean-matrix entries

Each entry receives an independent Gamma prior in shape–rate form,

$$M_{zz'} \sim \text{Gamma}(a_z, b_z),$$

with row-specific shape a_z and rate b_z .

Remark (Support restrictions are an implementation detail). In practice the implementation may exclude biologically implausible transitions by fixing the corresponding entries of M to zero (placing the Gamma prior only on the retained entries of each row). This sparsifies M but does not change the form of the model; sums over destination states below are read over the retained entries. We treat the support as a fixed preprocessing choice.

For the likelihood to support the data, the support guard is: for every non-missing destination state z' with $y_j(z') > 0$, at least one source state in the support of $\tilde{x}_j$ must reach z' (i.e. have $M_{zz'} > 0$).

3.4 Posterior over mean matrices

After observing the data $\mathcal{D}$, inference targets the posterior $p(M \mid \mathcal{D})$, which captures uncertainty about the joint tissue–phenotype mean kernel after all clone-level forward observations. Posterior summaries include posterior means $\mathbb{E}[M_{zz'} \mid \mathcal{D}]$ and credible intervals for individual entries.

3.5 Posterior predictive destination counts

Because M is latent, the intensity $\mu_j = \tilde{x}_j M$ is also uncertain. Posterior draws of M induce replicated destination counts

$$y_j^{\text{rep}}(z') \sim \text{Poisson}(d_{j,s'} \mu_j(z'))$$

over destination states with $m_{j,s'} = 1$.

3.6 Backward attribution

The reverse question—which source state produced an observed destination cell—is a posterior attribution. For observation j ,

$$w_{jzz'} = \frac{\tilde{x}_j(z) M_{zz'}}{\sum_{\tilde{z} \in \mathcal{Z}} \tilde{x}_j(\tilde{z}) M_{\tilde{z}z'}} = \frac{\tilde{x}_j(z) M_{zz'}}{\mu_j(z')},$$

the same form as the forward intensity with source and destination roles reversed. Being a function of M , it is a posterior read-out: evaluated at the fitted $\hat{M}$, or per draw to carry uncertainty, it attributes observed destination cells back to their likely sources.

4 Full Generative Story

The full model is a model for the one-step expansion and redistribution of clone mass across tissue–phenotype states. It can be summarized as follows.

Generative model

The state space $\mathcal{Z}$ ($|\mathcal{Z}| = 33$) and the fixed inputs—each observation's depth-rescaled source $\tilde{x}_j$, the destination depths $d_{j,s'}$, and the missingness flags $m_{j,s'}$ —are given. The model generates the destination counts as follows.

1. **Mean-matrix entries.** For each source state z , draw the row independently (entrywise),

$$M_{z\cdot} \sim \text{Gamma}(a_z, b_z),$$

shared across all patients and steps.

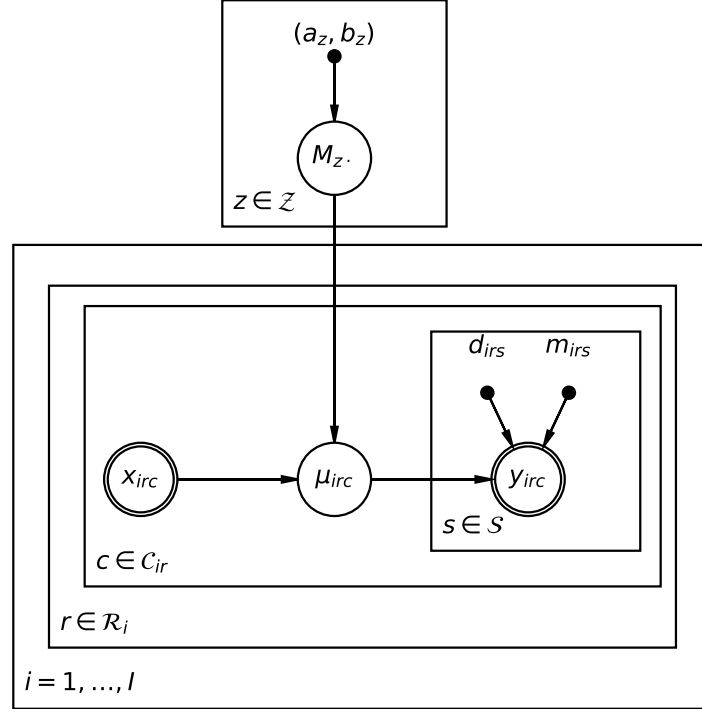
2. **Observation.** For each observation $j = (i, r, c)$, push the source forward to the intensity $\mu_j = \tilde{x}_j M$, and for each destination state z' with $m_{j,s'} = 1$, draw

$$y_j(z') \sim \text{Poisson}(d_{j,s'} \mu_j(z')), \quad s' = \text{tissue}(z').$$

Putting the pieces together, the generative model is, compactly,

$$M_{zz'} \sim \text{Gamma}(a_z, b_z), \quad \mu_j = \tilde{x}_j M, \quad y_j(z') \sim \text{Poisson}(d_{j,s'} \mu_j(z')) \quad (m_{j,s'} = 1),$$

with the corresponding dependency structure shown in Figure 1.



Node	Meaning
x_{irc}	observed source counts at the start of the step (over the 33 states)
$\tilde{x}_{irc} = x_{irc} / d_{irs}^{\text{src}}$	depth-rescaled source (deterministic; folded into the $x \rightarrow \mu$ edge)
$\mu_{irc} = \tilde{x}_{irc} M$	destination intensity vector
y_{irc}	observed destination counts in tissue s at the end of the step
M_z	latent mean-matrix row, shared across all patients and steps
d_{irs}	destination sequencing depth, Poisson exposure (fixed)
m_{irs}	sample missingness flag (fixed)
(a_z, b_z)	Gamma prior shape / rate (fixed)

Figure 1: Plate diagram of the generative model (node key above). The rows M_z are drawn once per source state z from a Gamma prior (a_z, b_z) ; this block sits *outside* the patient plate, so a single M is shared across all patients and steps. The observation side nests patient i , step $r \in \mathcal{R}_i$, clone $c \in \mathcal{C}_{ir}$, and tissue $s \in \mathcal{S}$. The depth-rescaled source $\tilde{x}_{irc}$ is pushed through M to the intensity $\mu_{irc} = \tilde{x}_{irc} M$, and each destination count y_{irc} is a Poisson draw with mean $d_{irs} \mu_{irc}$ set by its tissue's depth d_{irs} and gated by its missingness m_{irs} . Double circles (x_{irc} , y_{irc}) mark observed quantities; solid dots ((a_z, b_z) , d_{irs} , m_{irs}) mark fixed inputs; the depth-rescaling $\tilde{x}_{irc} = \text{Rescale}(x_{irc})$ is folded into the $x \rightarrow \mu$ edge.

5 Inference

Inference estimates the posterior over the shared mean matrix M . The observed data are the clone-level forward observations $\mathcal{D}$; the depth-rescaled sources $\tilde{x}_j$, the depths $d_{j,s'}$, and the missingness flags $m_{j,s'}$ are fixed inputs, as are the Gamma prior parameters a_z, b_z .

5.1 Learned quantity

The main learned object is the posterior $p(M \mid \mathcal{D})$. A single-entry summary is

$$\mathbb{E}[M_{(a,u),(b,v)} \mid \mathcal{D}],$$

the posterior mean number of destination- (b, v) cells produced per source- (a, u) cell over one forward step.

5.2 Posterior sampling

The posterior is not available in closed form. By Poisson superposition the per-source allocations of each destination count marginalize exactly, leaving M as the only unknown—a smooth target over its $L \times L$ entries, with no auxiliary allocation variables. We sample M directly from $p(M \mid \mathcal{D})$, drawing $M^{(1)}, \dots, M^{(S)}$ by gradient-based Markov chain Monte Carlo; the log-posterior is differentiable in M (the intensity $\mu_j = \tilde{x}_j M$ is linear in M , and the Gamma prior and Poisson likelihood are smooth on the positive orthant).

5.3 Estimated mean matrix and uncertainty

The fitted mean matrix is the posterior mean $\hat{M} = \mathbb{E}[M \mid \mathcal{D}]$, estimated by the sample average $\frac{1}{S} \sum_s M^{(s)}$; its row sums are free— $\hat{M}$ is not normalized across destinations. Credible intervals for any entry follow from the sample quantiles. Every read-out below is a deterministic function of M , so evaluating it on each draw $M^{(s)}$ yields its full posterior, summarized by a posterior mean and credible interval.

6 Posterior Summaries

The fitted $\hat{M}$ yields, for any source tissue and phenotype, three per-step quantities—where a clone’s mass traffics across tissues, whether it expands or contracts, and how its phenotype mix reshapes—each a dimensionless fraction or factor over one step, not a per-time rate.

Define $\rho^a(u)$ as the empirical source-cell phenotype distribution in tissue a , pooled across clones and steps; it collapses source phenotypes in the tissue-level rows below.

Here T and the per-state expansion factor g_z are the factors of $\hat{M} = \text{diag}(g) T$ from Section 2.1, and $\theta(z)$ is a clone’s source composition—its source cells as a distribution over states.

Quantity	Grain	Reads as	Expression
Traffic (re-solved)	(tissue, phenotype) $\rightarrow$ tissue	fraction of mass, per step	$\sum_v T_{(a,u),(b,v)}$
Traffic (collapsed)	tissue $\rightarrow$ tissue	fraction of mass, per step	$\sum_u \rho^a(u) \sum_v T_{(a,u),(b,v)}$
Traffic rate	tissue	retention vs. out, per step	stays = $(b=a)$, out = $\sum_{b \neq a}$; sum to 1
Expansion / contraction	clone	factor, per step (> 1 expand, < 1 contract, $\rightarrow 0$ clonal loss)	$g_z = \sum_{z'} \widehat{M}_{zz'}$; clone = $\sum_z \theta(z) g_z$
Switching (re-solved)	(tissue, phenotype) $\rightarrow$ tissue	phenotype distribution, per step	$h_{a,u \rightarrow b}(v) = \frac{T_{(a,u),(b,v)}}{\sum_{v'} T_{(a,u),(b,v')}}$
Switching (collapsed)	tissue $\rightarrow$ tissue	fraction of mass, per step	$\sum_u \rho^a(u) h_{a,u \rightarrow b}(v)$
Switch rate	tissue route	switched vs. kept, per step	switched = $\sum_{v \neq u} h$, kept = $h(u)$

Table 2: Per-step read-outs of the fitted mean matrix $\widehat{M}$.